SUVIN POOJARY

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PROFESSIONAL SUMMARY

Final-year Artificial Intelligence & Machine Learning undergraduate with experience developing AI systems across healthcare, large language models, and computer vision. Hands-on experience with LLM-based retrieval, RAG, vector databases, SQL-driven data retrieval, and AI-assisted healthcare applications. Currently developing a multimodal deep learning framework for kidney tumor risk assessment while exploring reliable and production-oriented AI systems.

RESEARCH INTERESTS

AI for Healthcare · Multimodal Learning · LLMs, NLP & RAG · Computer Vision & 3D Reconstruction · Reliable AI Systems

EDUCATION

NMAM Institute of Technology (Nitte), Udupi *August 2023 – Present*

Bachelor of Technology in Artificial Intelligence and Machine Learning

- CGPA: 8.46 / 10.0
- Coursework: Machine Learning, Artificial Intelligence, Cloud Computing, Cybersecurity, DSA

ACADEMIC RESEARCH

MetaFusion-Driven Multimodal Learning for Kidney Tumor Risk Assessment *Final Year Research Project*

- Developing a multimodal deep learning framework for kidney tumor risk stratification by combining abdominal CT imaging with clinical metadata from the KNIGHT Challenge dataset.
- Adapting the MetaFusion architecture with a ResNet-18 image backbone and clinical feature representation to perform both five-way risk classification and binary high/very-high-risk classification.
- Preprocessing 3D CT volumes using tumor segmentation masks, extracting the axial slice with the largest tumor cross-sectional area, and applying soft-tissue CT windowing for model input.

EXPERIENCE

Student Intern — Center for Software Development, MAHE, Manipal *Jun 2026 – Aug 2026*

- Developed and integrated prompt-based natural-language data retrieval functionality for a healthcare application using LLM, SQL-based retrieval, and backend APIs.
- Worked with FastAPI, Qdrant, Docker, and database connectivity to support retrieval and reporting workflows.
- Implemented and tested automated healthcare reporting and data-retrieval functionality while debugging backend and integration issues.
- Followed institutional confidentiality and information-security practices while working with sensitive healthcare data.

PROJECTS

PharmaGuard — AI-Assisted Pharmacovigilance Platform

- Built an AI-assisted drug-safety decision-support platform that ingests adverse drug reaction (ADR) reports and routes AI-generated outputs through a pharmacist approval workflow with a full audit trail.
- Designed a multi-stage, traceable LLM pipeline (de-identification, extraction, classification, RAG-based evidence retrieval, summarization) allowing step-by-step debugging.
- Implemented a PII de-identification module and a RAG retrieval layer (Sentence-Transformers + Qdrant vector search) surfacing past cases and drug-label evidence with similarity scores and citations.
- Architected as three independently deployable containerized services (React client, Express API, FastAPI ML microservice).

HoloGen 3D-Reconstructor

- Developed a model to convert 2D images into 3D scenes using PyTorch, OpenCV, and PyRender.
- Focused on monocular depth estimation for accurate 3D reconstruction and visualization.
- Implemented image preprocessing and depth estimation workflows to generate depth information from single RGB images.
- Generated and visualized reconstructed 3D scenes using estimated depth maps, enabling spatial representation of objects from 2D inputs.

WaveHead

- Built a team-based music player app with MySQL database integration for managing users and playlists.
- Designed an intuitive UI and ensured seamless audio playback.

Sketch Predictor

- Created a browser-based sketch recognition tool using Svelte, Tailwind CSS, and a decision tree model trained on the QuickDraw dataset.
- Deployed client-side for instant, offline predictions.

SKILLS

- **Programming:** Python, C++, Java, SQL
- **AI & Machine Learning:** Deep Learning, Machine Learning, Computer Vision, NLP, Predictive Modeling, LLMs, RAG
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, U-Net, PyRender, Pandas, NumPy
- **AI Systems & Backend:** FastAPI, Qdrant, Docker, Node.js, React
- **Tools:** Git/GitHub, Jupyter, Google Colab, VS Code
- **Languages:** Fluency in English, Hindi, Kannada

CERTIFICATIONS

- Google Cloud Computing Foundations — NPTEL (Aug – Oct 2024)
- Summer Research Internship in “Machine Learning for Medical and Pharmaceutical Sciences” — NMAM Institute of Technology (May 2024)